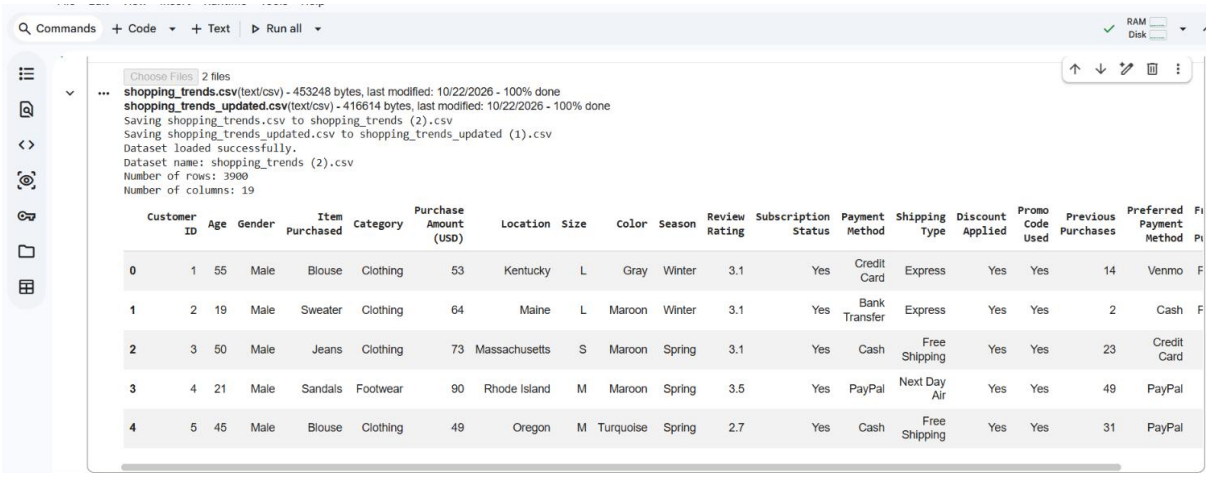
1. Introduction

Data cleaning and exploratory data analysis are important steps in the data analysis process. Data cleaning improves data quality by identifying missing values, duplicate records, and extreme observations. Exploratory Data Analysis (EDA) helps understand the distribution of variables, relationships between customer characteristics, and shopping behaviour.

This assignment uses the **Customer Shopping Trends** dataset to apply data cleaning techniques, categorical encoding, normalization, standardization, and data visualization using Google Colab. The analysis aims to identify meaningful patterns in customer purchasing behaviour and understand how these findings can support business decisions.

2. Dataset description

Item	What to report
Dataset name	Customer Shopping Trends
Source	Kaggle
Domain	E-commerce / Marketing
Number of rows	3900
Number of columns	19



	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Payment Method	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Preferred Payment Method
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Credit Card	Express	Yes	Yes	14	Venmo
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Bank Transfer	Express	Yes	Yes	2	Cash
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Cash	Free Shipping	Yes	Yes	23	Credit Card
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	PayPal	Next Day Air	Yes	Yes	49	PayPal
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Cash	Free Shipping	Yes	Yes	31	PayPal

3.Data cleaning

3.1 Handling missing values

Missing values were identified using Pandas. The number and percentage of missing observations were calculated for each column. Median imputation was used for numerical variables because the median is less affected by extreme values. Mode imputation was used for categorical variables. If no valid category was available, the value was marked as "Unknown".

season	0
review_rating	0
subscription_status	0
payment_method	0
shipping_type	0
discount_applied	0
promo_code_used	0
previous_purchases	0
preferred_payment_method	0
frequency_of_purchases	0

```
Total remaining missing values: 0
```

3.2 Duplicate removal

Duplicate records were checked using the duplicated() function.

Exact duplicate rows were removed using drop_duplicates() to prevent repeated records from affecting the analysis.

```
print("Duplicate rows before cleaning:", df.duplicated().sum())

df = df.drop_duplicates().copy()
df.reset_index(drop=True, inplace=True)

print("Duplicate rows after cleaning:", df.duplicated().sum())
print("Dataset shape after duplicate removal:", df.shape)
```

```
Duplicate rows before cleaning: 0
Duplicate rows after cleaning: 0
Dataset shape after duplicate removal: (3900, 19)
```

3.3 Outlier detection and treatment

Potential outliers were identified using the Interquartile Range (IQR) method. Boxplots were used to visualize the distribution of numerical variables. Observations below the lower IQR bound or above the upper IQR bound were flagged as potential outliers. Capping was applied to limit extreme values without deleting complete records.

```
outlier_summary = pd.DataFrame(outlier_results)
display(outlier_summary)
```

	Column	Q1	Q3	IQR	Lower_Bound	Upper_Bound	Outlier_Count
0	age	31.0	57.0	26.0	-8.00	96.00	0
1	purchase_amount_usd	39.0	81.0	42.0	-24.00	144.00	0
2	review_rating	3.1	4.4	1.3	1.15	6.35	0
3	previous_purchases	13.0	38.0	25.0	-24.50	75.50	0

Next steps: Generate code with outlier_summary | New interactive sheet

	Column	Outliers_Before	Values_Capped	Lower_Cap	Upper_Cap
0	age	0	0	-8.00	96.00
1	purchase_amount_usd	0	0	-24.00	144.00
2	review_rating	0	0	1.15	6.35
3	previous_purchases	0	0	-24.50	75.50

Shape after outlier treatment: (3900, 19)

3.4 Categorical encoding

One-Hot Encoding was applied to categorical features to convert them into numerical indicator columns. This makes categorical information suitable for numerical analysis and machine-learning algorithms. Identifier columns were excluded because they represent record identities rather than meaningful predictive features.

3.5 Normalization and standardization

Min-Max Scaling was applied to transform numerical features to a range between 0 and 1. Z-score Standardization was applied separately to transform numerical features to a scale with mean approximately zero and standard deviation approximately one. These transformations help place numerical variables on comparable scales.

Min-Max normalized columns:
['age', 'purchase_amount_usd', 'review_rating', 'previous_purchases']

	age	purchase_amount_usd	review_rating	previous_purchases
0	0.711538	0.4125	0.24	0.265306
1	0.019231	0.5500	0.24	0.020408
2	0.615385	0.6625	0.24	0.448980
3	0.057692	0.8750	0.40	0.979592
4	0.519231	0.3625	0.08	0.612245

Minimum values after scaling:

	0
age	0.0
purchase_amount_usd	0.0
review_rating	0.0
previous_purchases	0.0

dtype: float64

Maximum values after scaling:

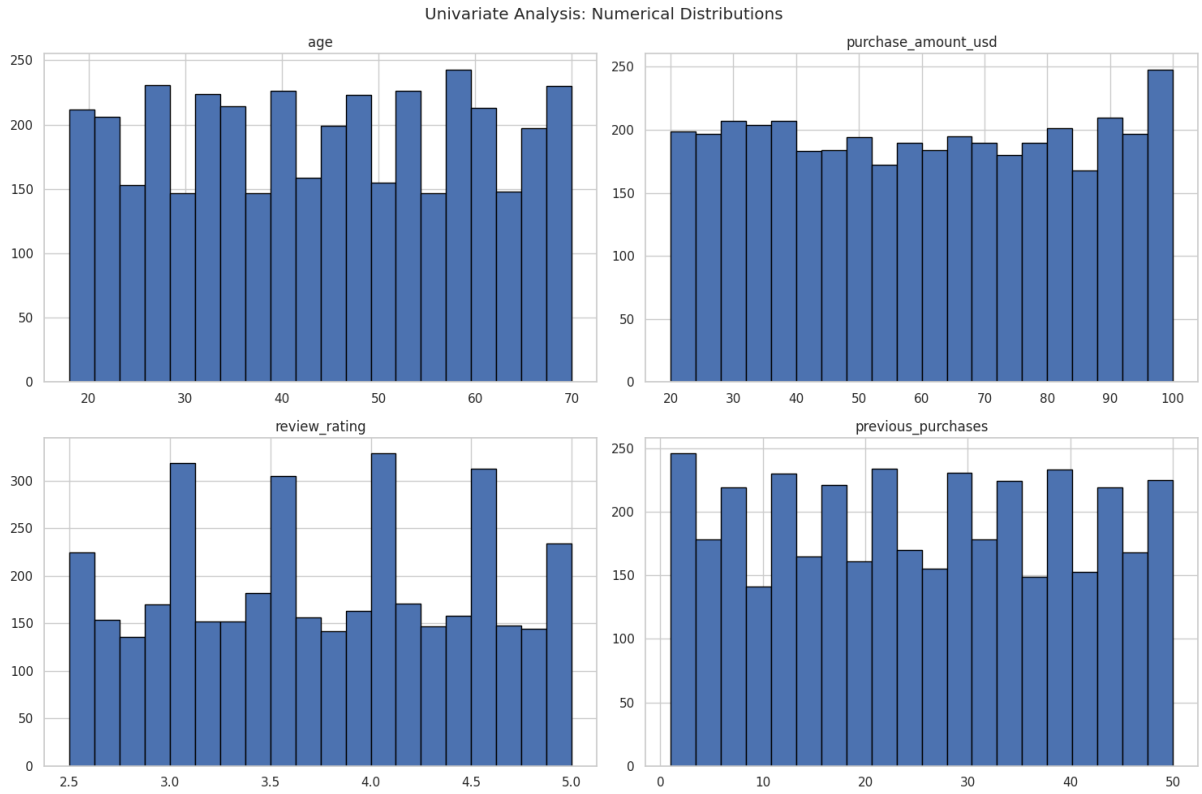
	0
age	1.0
purchase_amount_usd	1.0
review_rating	1.0
previous_purchases	1.0

dtype: float64

4.Exploratory Data Analysis

4.1 Univariate analysis

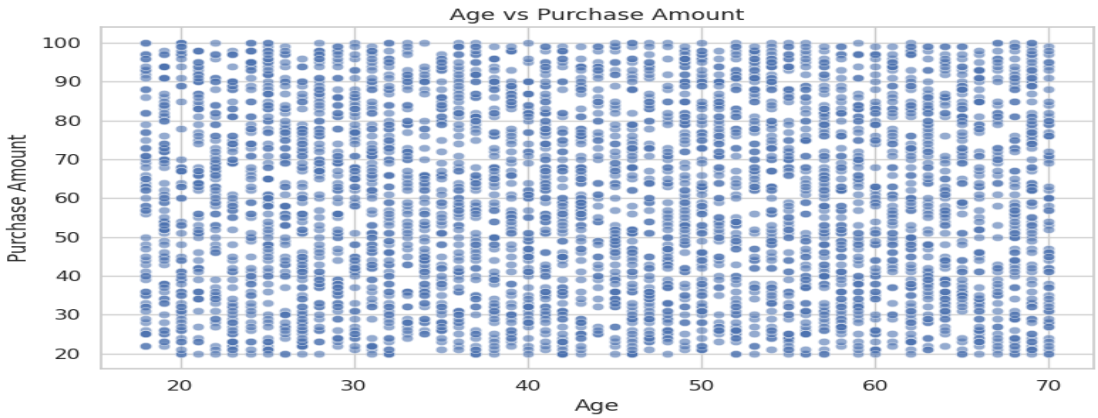
Univariate analysis was conducted to understand individual variables. Histograms were used to examine the distribution of numerical variables, bar charts were used to compare categorical frequencies, and boxplots were used to examine spread and potential outliers.



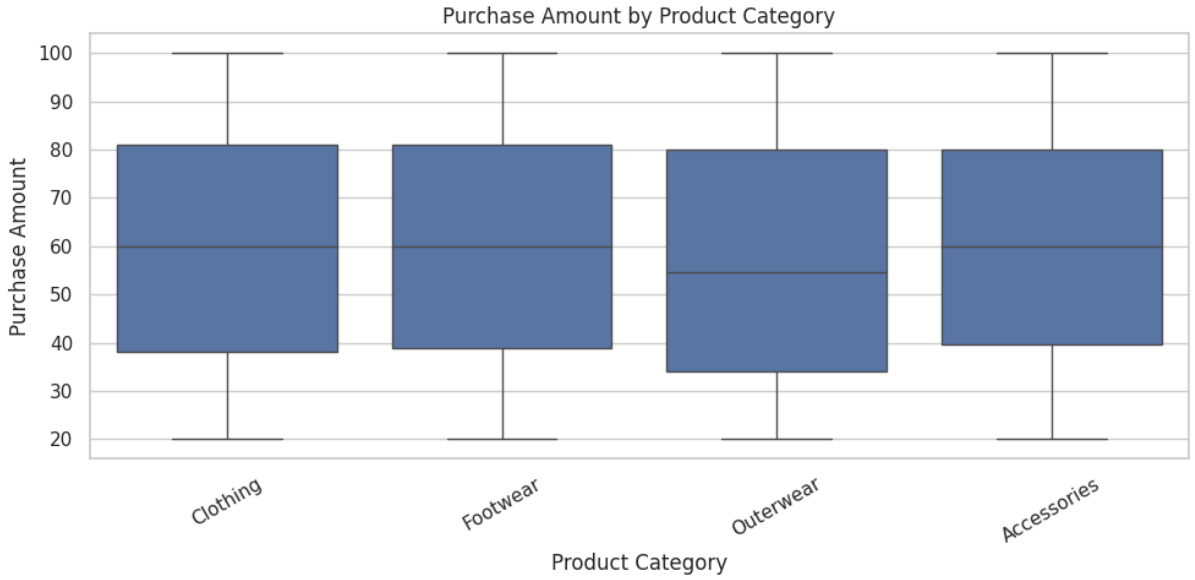
4.2 Bivariate analysis

Bivariate analysis was used to explore relationships between two variables. A scatter plot was created to examine age and purchase amount. Boxplots were used to compare purchase amounts across product categories, and a bar chart compared average purchase amounts across gender groups.

Scatter Plot :

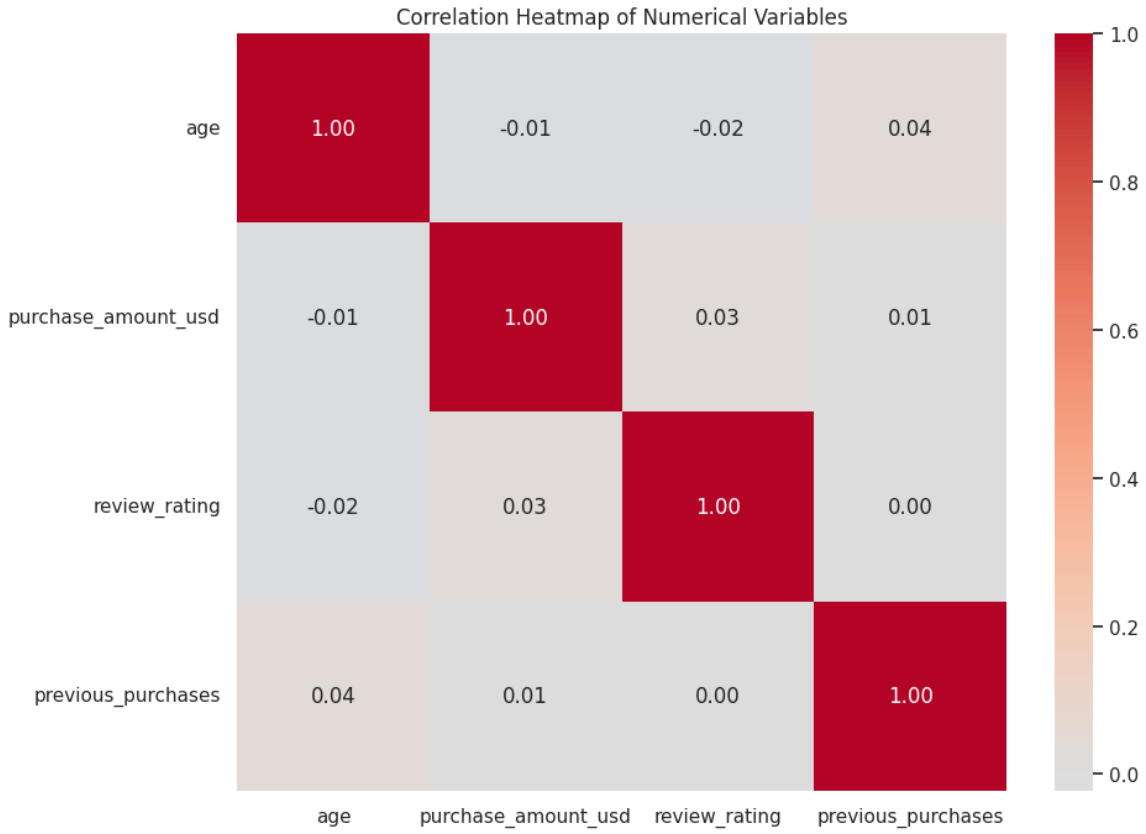


Bar Chart:



4.3 Multivariate analysis

Multivariate analysis was performed using a correlation heatmap, pair plot, and grouped bar chart. The heatmap summarizes the direction and strength of linear relationships between numerical variables. The pair plot provides a visual comparison of selected numerical variables, while the grouped bar chart compares average purchase amounts across product categories and gender groups.



4.4 Key insights from the actual data:

1. Most frequent category: Clothing (1737 records).
2. Highest average purchase amount by category: Footwear (60.26).
3. Highest average purchase amount by gender: Female (60.25).
4. Strongest absolute numerical correlation: age and previous_purchases (r = 0.040). Correlation does not establish causation.
5. Records marked as subscribed: 27.00%.